

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	<i>Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)</i>		
Student Name:	G. Tejasree	Reg. No.:	192511204
Section / Batch:	Section A	Date:	30 August 2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

DATA INTERPRETATION

6. Structure from Motion Consistency

Given Data

Two datasets are used for Structure from Motion:

- **Dataset A:** Accurate depth but requires high-quality images.
- **Dataset B:** Inconsistent depth but works with low-quality images.

1. Understanding of Data

Structure from Motion (SfM) is a Computer Vision technique used to estimate the **3D structure of a scene and camera motion from a sequence of 2D images**. It identifies common features between different images and uses their changes in position to estimate depth and camera movement.

The two datasets have different advantages.

Dataset A provides accurate depth estimation, which means the estimated distance and 3D position of objects are more reliable. However, it requires high-quality images. High resolution, good lighting, sharp edges, and clear textures are generally helpful for accurate feature matching.

Dataset B has an advantage in terms of image-quality requirements. It can work with low-quality images, making it more robust to noisy, blurred, or low-resolution input. However, its depth estimation is inconsistent. Inconsistent depth can produce errors in the reconstructed 3D scene.

2. Data Analysis

The main comparison is between **depth accuracy** and **robustness to image quality**.

Factor	Dataset A	Dataset B
Depth accuracy	High	Lower/Inconsistent
Depth consistency	High	Low
Image quality requirement	High	Low
Robustness to poor images	Lower	Higher
3D reconstruction quality	Better	Less reliable
Real-world flexibility	Moderate	High

Dataset A is better when the input images are controlled and high quality. Accurate depth is particularly important because small depth errors can accumulate during 3D reconstruction.

Dataset B can operate in more difficult imaging conditions. However, if the depth values change inconsistently between frames, the reconstructed scene may contain incorrect object positions or distorted geometry.

Therefore, there is a trade-off:

Dataset A → Accuracy and consistency

Dataset B → Robustness to poor-quality input

3. Application of Concepts

Dataset A is suitable for applications such as:

- 3D mapping
- Robot navigation
- Augmented Reality
- 3D reconstruction
- Autonomous navigation
- Industrial inspection

For example, in robot navigation, incorrect depth could cause the robot to misunderstand the distance to an obstacle. Therefore, consistent depth is highly important.

Dataset B may be useful for:

- Surveillance systems
- Low-cost cameras
- Outdoor environments
- Systems where image quality cannot be controlled
- Applications where approximate depth is acceptable

If the application needs highly reliable 3D information, Dataset A should be selected.

4. Evaluation

For real-world deployment, **Dataset A is generally the better choice when accurate depth is the primary requirement**. Although it requires high-quality images, the resulting depth information is more consistent and reliable.

However, Dataset B could be preferred when the system frequently receives low-quality images and maintaining operation is more important than achieving highly accurate depth.

5. Conclusion

Dataset A is more suitable for accuracy-critical real-world applications because it provides accurate and consistent depth. Dataset B provides better robustness to low-quality images but suffers from inconsistent depth.

Therefore, the final selection depends on the application:

Accuracy-critical application → Dataset A

Poor-image/low-quality environment → Dataset B

7. Stereo Vision vs Motion Estimation

Given Data

The depth estimation results are:

- **Stereo Vision:** Accurate but requires dual cameras.
- **Motion Estimation:** Less accurate but works with a single camera.

1. Understanding of Data

Stereo Vision is a depth estimation technique that uses two cameras positioned at different viewpoints. The same object appears at slightly different positions in the two images. This difference, called **disparity**, is used to calculate the object's depth.

Motion Estimation, also called monocular motion-based estimation in this context, uses a single camera and estimates depth from changes in the scene across multiple frames. It can determine information about camera movement and relative scene structure.

The main difference is therefore:

- Stereo Vision requires **two cameras** and provides better depth accuracy.
- Motion Estimation requires **one camera** but generally provides less accurate depth.

2. Data Analysis

Parameter	Stereo Vision	Motion Estimation
Number of cameras	2	1
Depth accuracy	High	Moderate
Hardware cost	Higher	Lower
Calibration	Required	Less hardware calibration
Computational complexity	Moderate/High	Moderate
Hardware requirement	Higher	Lower
Suitable for low-cost systems	Less suitable	More suitable

Stereo Vision has the advantage of directly obtaining depth from the difference between two camera views. However, the two cameras must be properly aligned and calibrated.

Motion Estimation has a major hardware advantage because only one camera is required. This reduces system cost, size, and power requirements.

However, motion estimation can be affected by:

- Camera movement
- Scene texture
- Lighting changes
- Motion blur

- Lack of visual features
- Errors in tracking between frames

3. Application of Concepts

Stereo Vision is useful in:

- Autonomous vehicles
- Industrial robots
- 3D measurement
- Robotic navigation
- Object distance measurement

For example, an autonomous vehicle may require accurate distance information to identify obstacles.

Motion Estimation is useful in:

- Smartphones
- Drones
- Mobile robots
- Low-cost vision systems
- Applications where only one camera is available

4. Evaluation Under Hardware Constraints

When hardware resources are limited, adding a second camera increases:

- Hardware cost
- Physical space requirements
- Power consumption
- Calibration complexity
- System design complexity

Therefore, if the application has a strict hardware or cost limitation, Motion Estimation becomes more practical.

If the system has sufficient hardware and requires highly accurate depth, Stereo Vision is preferable.

5. Conclusion

Under **hardware constraints**, **Motion Estimation is preferable** because it requires only a single camera and reduces cost and system complexity.

However, when **depth accuracy is more important than hardware cost**, Stereo Vision should be selected.

Limited hardware → Motion Estimation

High accuracy requirement → Stereo Vision

8. Model Generalization vs Overfitting

Given Data

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

1. Understanding of Data

Training accuracy represents how well a model performs on the data used to train it.

Testing accuracy represents how well the model performs on **unseen data**.

The two models show different behavior.

Model A:

- Training accuracy = 98%
- Testing accuracy = 70%

Model B:

- Training accuracy = 90%
- Testing accuracy = 88%

At first glance, Model A appears better because it has 98% training accuracy. However, training accuracy alone does not determine whether a model is suitable for deployment.

2. Data Analysis

For Model A:

Training–Testing gap

$$= 98\% - 70\%$$

$$= \mathbf{28\%}$$

This is a very large gap.

It suggests that Model A has learned the training data extremely well but cannot generalize effectively to unseen data. This behavior is known as **overfitting**.

For Model B:

Training–Testing gap

$$= 90\% - 88\%$$

$$= \mathbf{2\%}$$

This is a very small difference.

It indicates that Model B performs consistently on both training and testing data. Therefore, Model B has better **generalization ability**.

3. Comparison

Factor	Model A	Model B
Training accuracy	98%	90%
Testing accuracy	70%	88%
Difference	28%	2%
Overfitting risk	High	Low
Generalization	Poor	Good
Unseen-data performance	Poorer	Better
Deployment reliability	Lower	Higher

The important observation is that **Model B has 18 percentage points higher testing accuracy than Model A:**

$$88\% - 70\% = 18\%$$

Therefore, Model B is clearly more reliable for unseen data.

4. Application of Concepts

In Computer Vision applications such as:

- Face recognition
- Object detection
- Medical image classification
- Image classification
- Autonomous systems

the model must work with new images that were not included during training.

If Model A is deployed, its high training accuracy may give a false impression of good performance.

Its 70% testing accuracy indicates that it may make many errors on new images.

Model B is more suitable because its training and testing performances are close.

5. Evaluation

The main objective of a machine learning model is not simply to memorize the training data. It should **learn general patterns that can be applied to new data**.

Therefore, testing accuracy and the difference between training and testing performance are important indicators.

Model A has:

High training accuracy + Low testing accuracy = Overfitting

Model B has:

Good training accuracy + High testing accuracy = Better generalization

6. Conclusion

Model B is more reliable for real-world deployment.

Although Model A has a higher training accuracy of 98%, its testing accuracy is only 70%, indicating a high risk of overfitting.

Model B achieves 88% testing accuracy with only a 2% training-testing gap. Hence, it has better generalization and is expected to perform more consistently on unseen real-world data.

9. YOLO Performance Across Scales

Given Data

YOLO detection results:

- **Large objects = 95%**
- **Medium objects = 85%**
- **Small objects = 60%**

1. Understanding of Data

YOLO (You Only Look Once) is an object detection algorithm that identifies objects and their locations in an image.

The given data shows that YOLO's detection accuracy changes according to object size.

- Large objects → **95%**
- Medium objects → **85%**
- Small objects → **60%**

Therefore, the detection performance decreases as object size becomes smaller.

2. Data Analysis

The performance difference between large and medium objects is:

$95\% - 85\% = 10\%$

The difference between medium and small objects is:

$85\% - 60\% = 25\%$

The overall difference between large and small objects is:

$95\% - 60\% = 35\%$

This is a significant difference.

Object Size	Accuracy	Performance
Large	95%	Excellent
Medium	85%	Good
Small	60%	Poor

The results show that YOLO performs very well when objects occupy a large portion of the image. Small objects are more difficult to detect because they contain fewer pixels and therefore provide less visual information.

For example, a nearby car may occupy hundreds of pixels, while a distant pedestrian may occupy only a small number of pixels. Important features of the small object can be lost during image resizing and feature extraction.

3. Application of Concepts

YOLO is widely useful for:

- Traffic monitoring
- Surveillance
- Autonomous vehicles
- Robotics
- Pedestrian detection
- Industrial inspection

For large objects, the given 95% accuracy indicates strong performance.

However, applications involving distant or tiny objects require better small-object detection.

Possible improvements include:

1. **Increasing image resolution**

Higher-resolution input provides more pixels for small objects.

2. **Multi-scale feature extraction**

Features from different image scales can be combined.

3. **Feature Pyramid Networks (FPN)**

FPN can improve detection of objects appearing at different scales.

4. **Data augmentation**

Training images can include different object sizes, positions, and conditions.

5. **More small-object training samples**

Including sufficient small-object examples can improve model learning.

6. **Improved detection architecture**

Multi-scale detection heads and better feature representations can improve small-object performance.

4. Evaluation

The data clearly indicates that YOLO's performance is **scale-dependent**.

The 95% accuracy for large objects is excellent, while the 60% accuracy for small objects indicates a major weakness.

Therefore, using standard YOLO alone may not be sufficient for applications where objects can appear at very different scales.

5. Conclusion

YOLO performs strongly on large objects with **95% accuracy** and reasonably well on medium objects with **85% accuracy**. However, its **60% accuracy for small objects** indicates a significant limitation.

Therefore, **YOLO alone is not sufficient for highly reliable multi-scale detection**. Multi-scale feature extraction, higher resolution, data augmentation, and improved detection strategies should be incorporated to improve small-object detection.

10. Motion Model Selection for Animation

Given Data

Two motion models are available:

- **Model A:** Smooth but physically inaccurate.
- **Model B:** Realistic but computationally intensive.

The models must be evaluated for different animation requirements, particularly **real-time systems and cinematic applications**.

1. Understanding of Data

A motion model determines how objects or characters move within an animation.

Model A produces smooth motion with lower computational requirements. However, it does not accurately follow physical laws. For example, acceleration, gravity, collisions, or natural movement may not be represented accurately.

Model B produces realistic and physically accurate motion. However, its computational requirements are high, which means it may require more processing power and time.

2. Data Analysis

Factor	Model A	Model B
Smoothness	High	High
Physical accuracy	Low	High
Realism	Moderate	High
Computational requirement	Low	High
Processing speed	Faster	Slower
Real-time suitability	High	Lower

Cinematic suitability	Moderate	High
------------------------------	----------	------

Model A prioritizes **performance and responsiveness**.

Model B prioritizes **realism and physical accuracy**.

Therefore, neither model is universally better. The appropriate model depends on the requirements of the animation system.

3. Application: Real-Time Animation

Real-time systems need to generate frames quickly.

Examples include:

- Video games
- Virtual Reality
- Interactive simulations
- Real-time character animation
- Augmented Reality

In these applications, computational resources are limited and delays can affect the user experience.

Therefore, **Model A is more suitable** because its lower computational requirement allows smooth and fast animation.

Even though it may not be physically perfect, smooth real-time performance is often more important.

4. Application: Cinematic Animation

Cinematic animation includes:

- Movies
- Animated films
- High-quality visual effects
- Offline 3D rendering

In these applications, rendering can take more time because the final output does not necessarily need to be generated instantly.

Realistic physical motion is more important than immediate response.

Therefore, **Model B is preferable** because it produces more realistic and physically accurate animation.

5. Evaluation

The selection can be represented as:

Real-time constraint → Model A

Because:

- Lower computation
- Faster processing

- Smooth performance
- Better responsiveness

Cinematic quality requirement → Model B

Because:

- Higher realism
- Better physical accuracy
- More natural movement
- Suitable for offline rendering

6. Conclusion

There is no single best motion model for every animation system.

Model A should be selected for real-time animation where speed, smoothness, and computational efficiency are important.

Model B should be selected for cinematic animation where physical accuracy and realistic movement are more important than processing time.

Thus, the decision should be based on the application's **performance constraints, computational resources, and required level of realism.**